

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 14

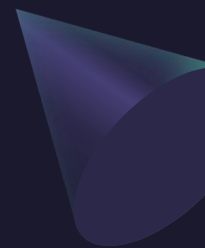


Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!
- Quiz on Friday on polar coordinates and beginning of triple integrals
- Exam I – One-Page Correction (+5) – due 3/9 – can be found on HuskyCT > Quizzes and Exams (will not be accepted late)



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

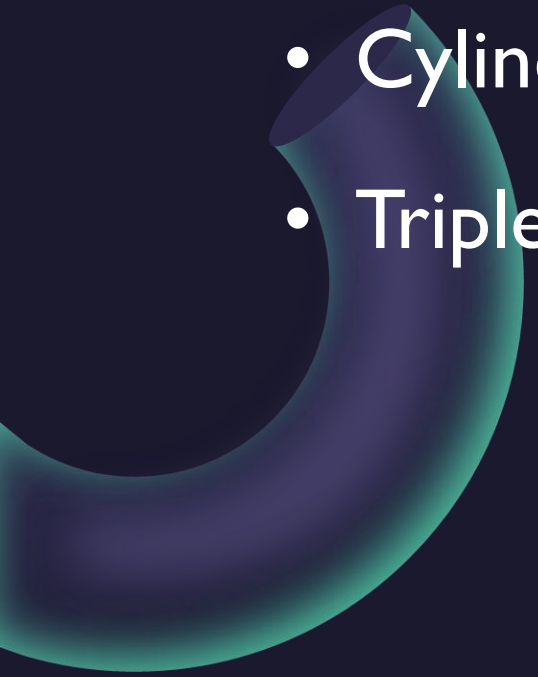
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More on Triple Integrals in Cylindrical Coordinates





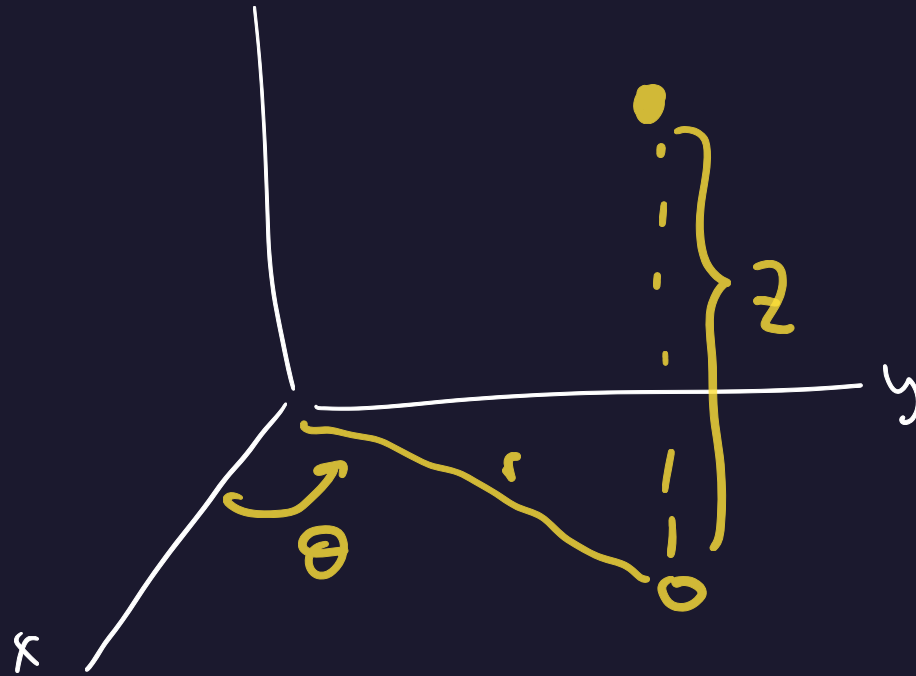
Today – Triple Integrals in Cylindrical Coordinates

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- Cylindrical Coordinates
 - Triple Integrals in Cylindrical Coordinates

Cylindrical Coordinates

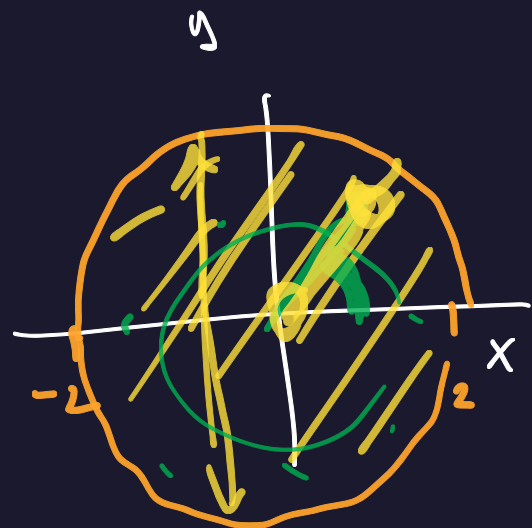
$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$



Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



$$y = \sqrt{4-x^2}$$

$$y^2 = 4-x^2$$

$$x^2 + y^2 = 4$$

$$x^2 + y^2 = r^2$$

$$\Rightarrow z = r$$

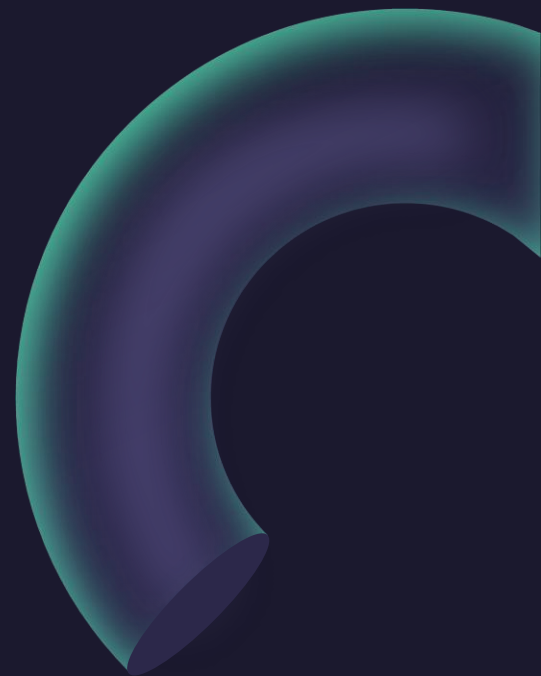
$$z = 2$$

$$z = \sqrt{x^2 + y^2} \leq 2$$

$$x^2 + y^2 = 4$$



$$\begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq 2 \\ r \leq z \leq 2 \end{cases}$$



Example: Evaluate the value of the triple integral $\int \int \int r^2$

$$I = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) dz dy dx$$

$$I = \int_0^{2\pi} \int_0^2 \int_r^2 r^2 \cdot r \cdot dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^2 r^3 z \Big|_r^2 dr d\theta = \int_0^{2\pi} \int_0^2 r^3 (2-r) dr d\theta$$

$$= \int_0^{2\pi} \left(\frac{2r^4}{4} - \frac{r^5}{5} \right) \Big|_0^2 d\theta = \int_0^{2\pi} \frac{16}{10} d\theta = \frac{16}{10} \theta \Big|_0^{2\pi} = \frac{16}{10} \cdot 2\pi$$

$$= \boxed{\frac{16}{5} \pi}$$

Example: Evaluate the value of the triple integral

$$I = \iiint_E \sqrt{x^2 + z^2} \, dV$$

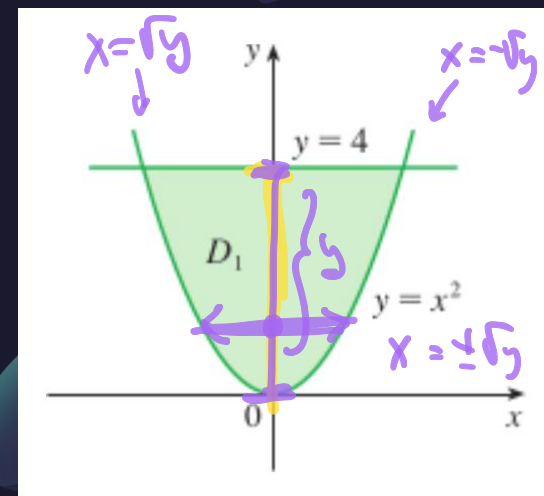
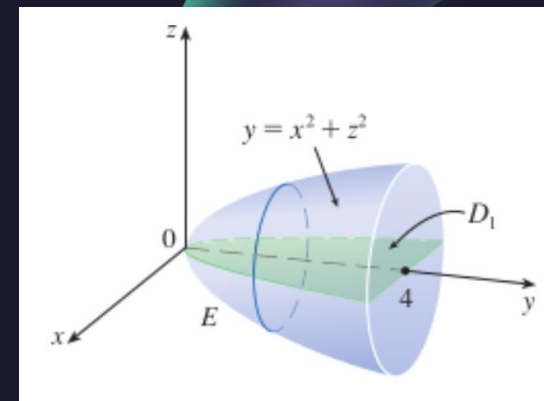
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

$$I = \int_0^4 \int_{-\sqrt{y}}^{\sqrt{y}} \int_{-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2 + z^2} \, dz \, dx \, dy$$



$$z^2 = y - x^2$$

$$z = \pm \sqrt{y - x^2}$$



Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

$$x = r \cos \theta$$

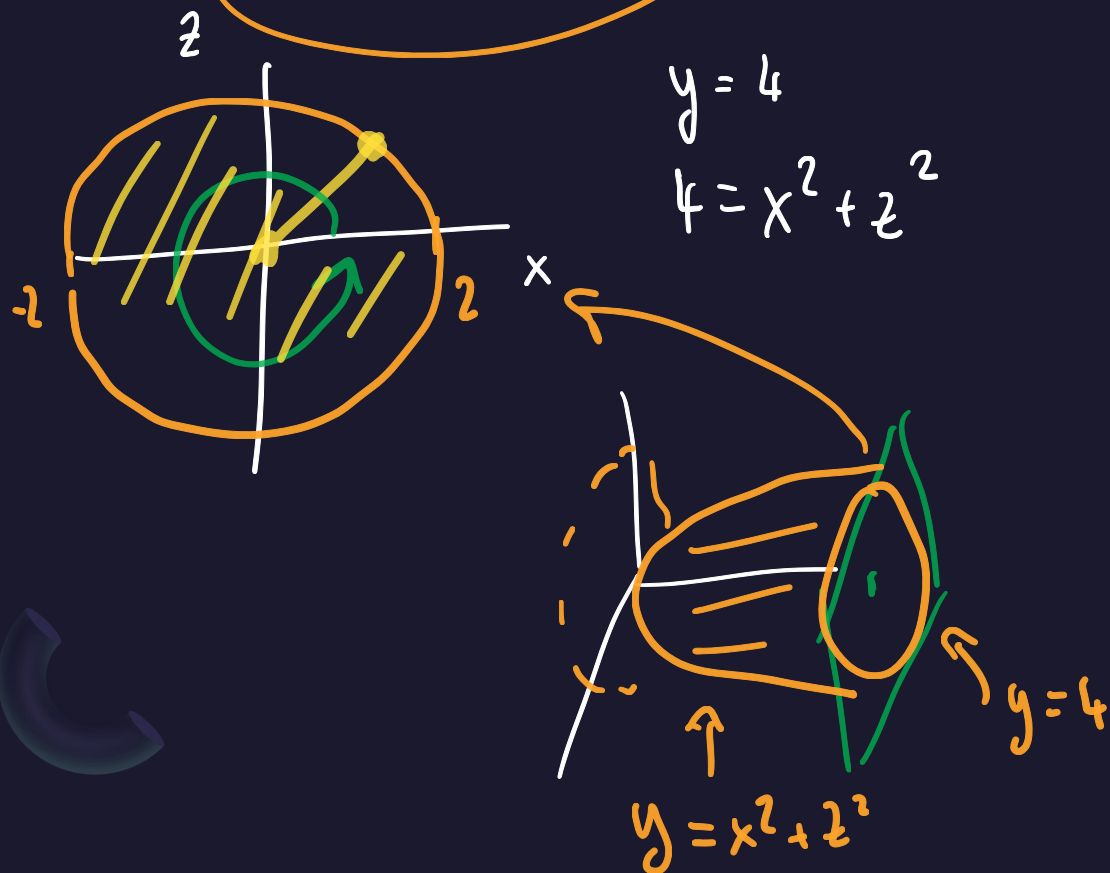
$$z = r \sin \theta$$

$$x^2 + z^2 = r^2$$

Cylindrical coords wrt y -axis.

$$\begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq 2 \\ x^2 + z^2 \leq y \leq 4 \end{cases} \rightarrow r^2 \leq y \leq 4$$

$$\int_0^{2\pi} \int_0^2 \int_{r^2}^4 r \cdot r \, dy \, dr \, d\theta = \boxed{\frac{128}{15} \pi}$$



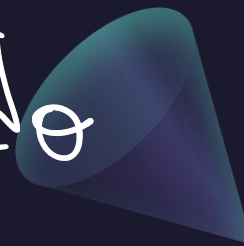
Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



$$\begin{cases} 0 \leq \theta \leq \pi \\ 0 \leq r \leq 1 \\ 1 - x^2 - y^2 \leq z \leq 4 \end{cases}$$

$$\Rightarrow \begin{cases} 0 \leq \theta \leq \pi \\ 0 \leq r \leq 1 \\ 1 - r^2 \leq z \leq 4 \end{cases}$$

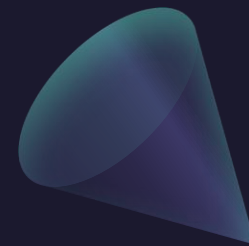
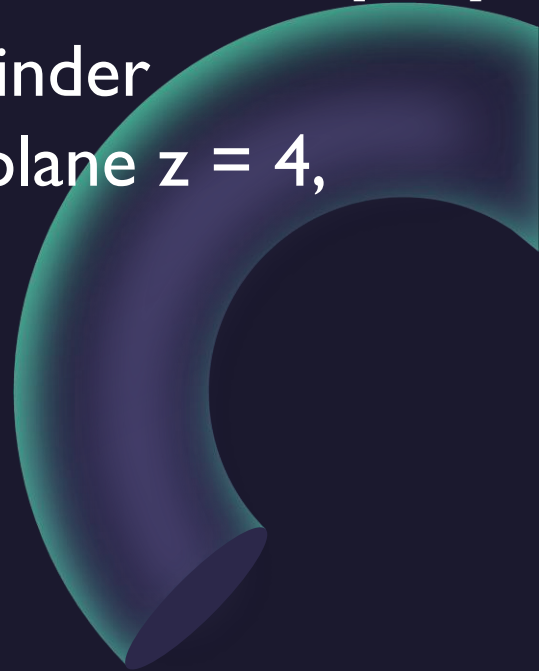
$$\int_0^{\pi} \int_0^1 \int_{1-r^2}^4 1 \cdot r \cdot dz \, dr \, d\theta$$



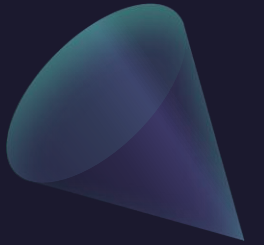
Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.

$$\int_0^{\pi} \int_0^1 r(4 - (1 - r^2)) \, dr \, d\theta$$

$$\int_0^{\pi} \int_0^1 3r + r^3 \, dr \, d\theta = \dots = \boxed{\frac{7}{4} \pi}$$

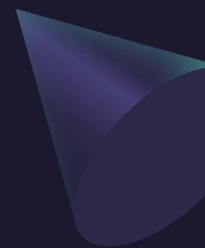


Questions?





ALVARO: Start the recording!



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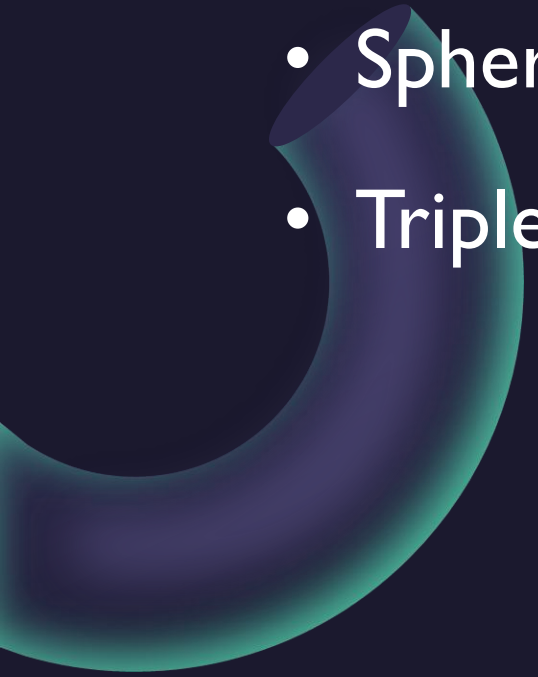
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Triple Integrals in Spherical Coordinates

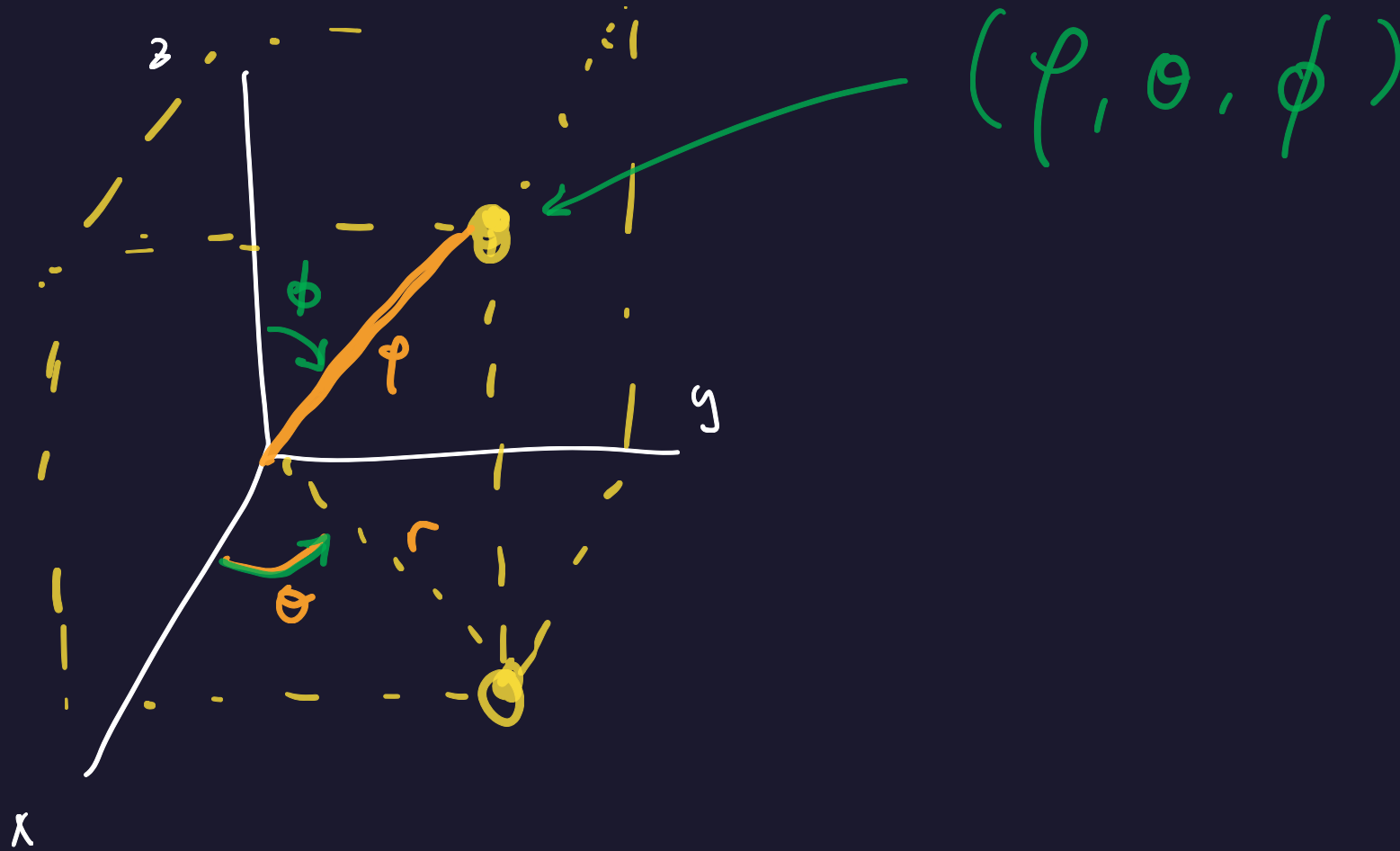


A small, dark teal sphere is positioned above a small, dark teal cube, both located to the left of the main title.

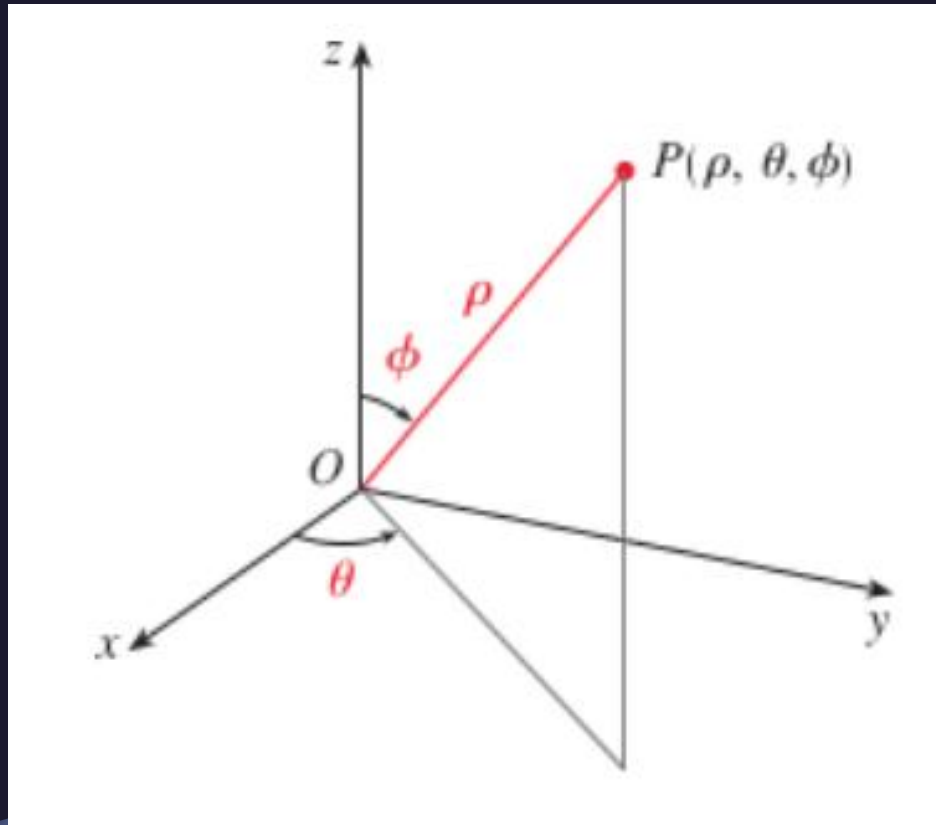
Today – Triple Integrals in Spherical Coordinates

- 
- A large, dark teal ring or torus is positioned on the left side of the slide, partially overlapping the list of topics.
- Spherical Coordinates
 - Triple Integrals in Spherical Coordinates

Spherical Coordinates



Spherical Coordinates



$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

$$\rho^2 = x^2 + y^2 + z^2$$

Cylindrical vs Spherical Coordinates

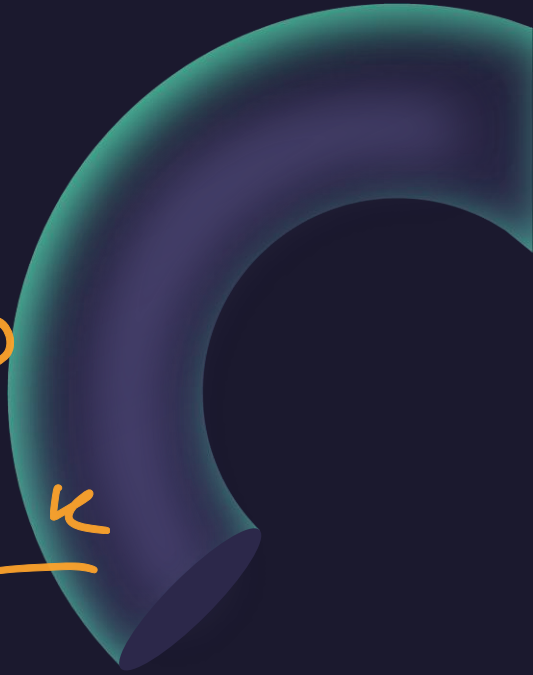
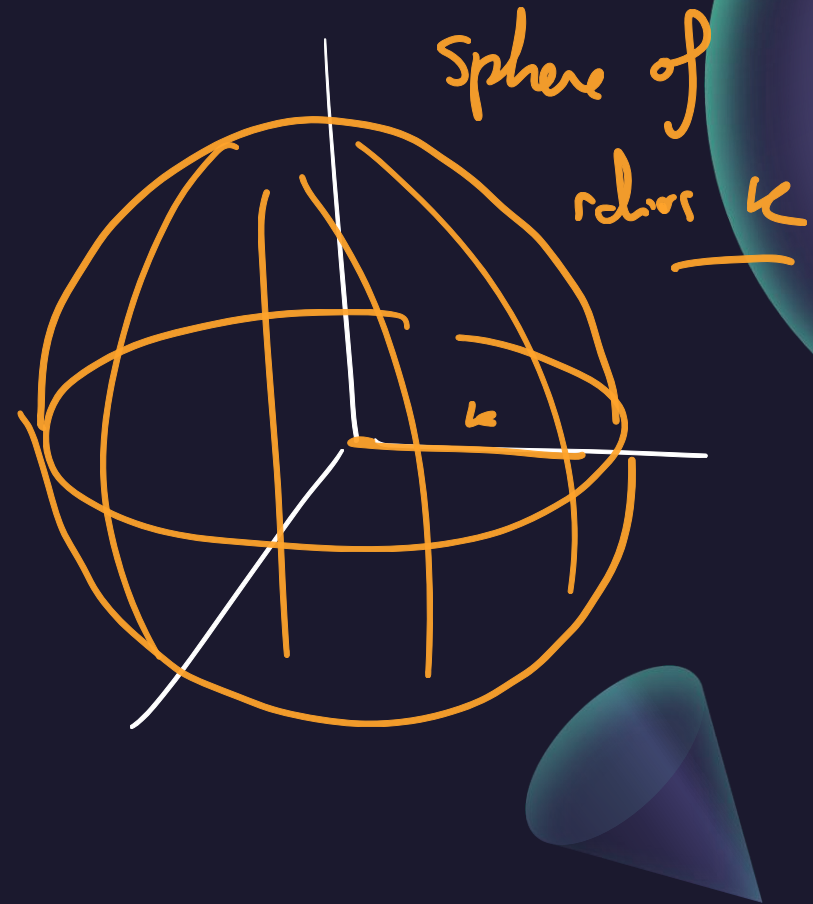
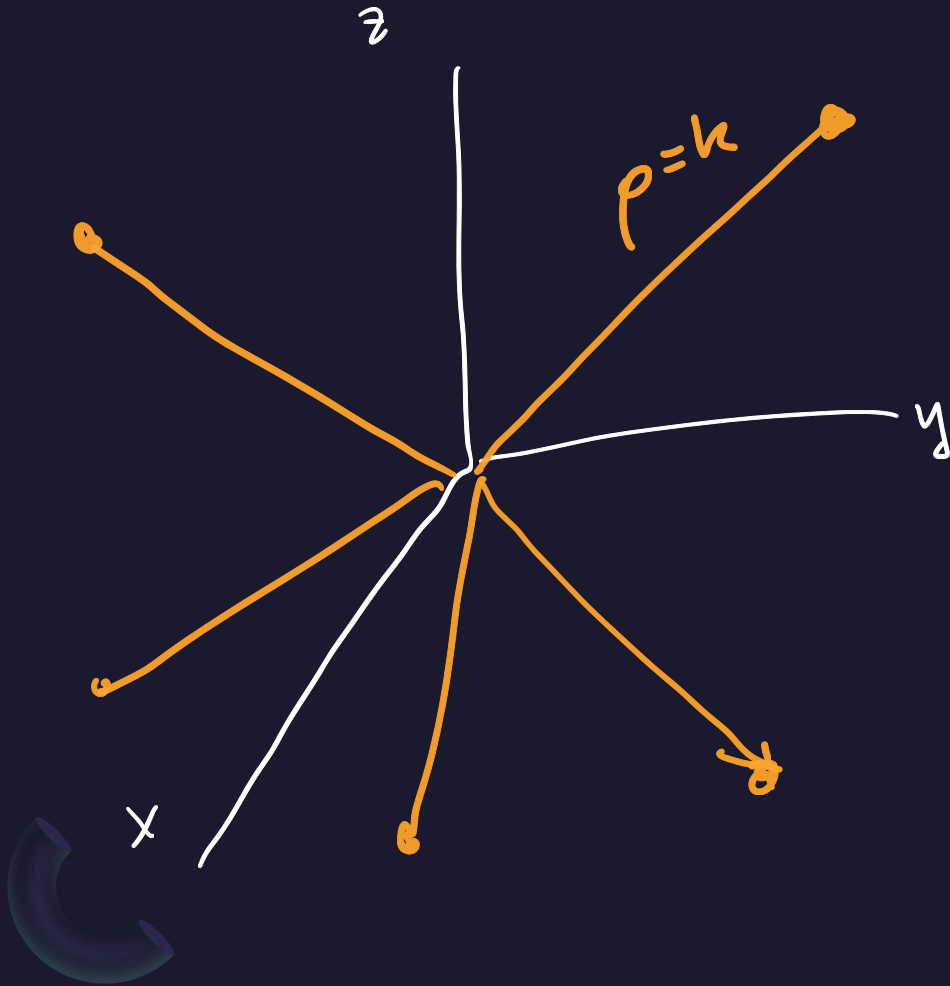
$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$

$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

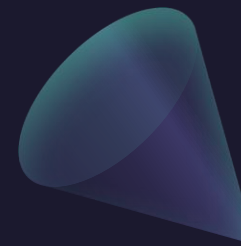
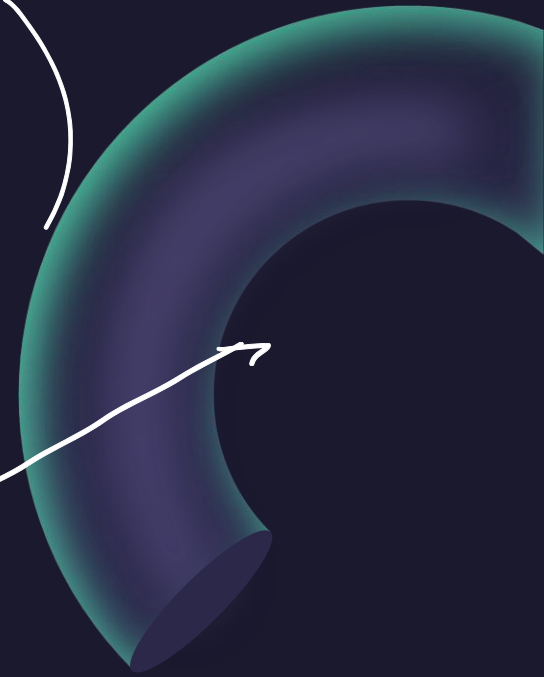
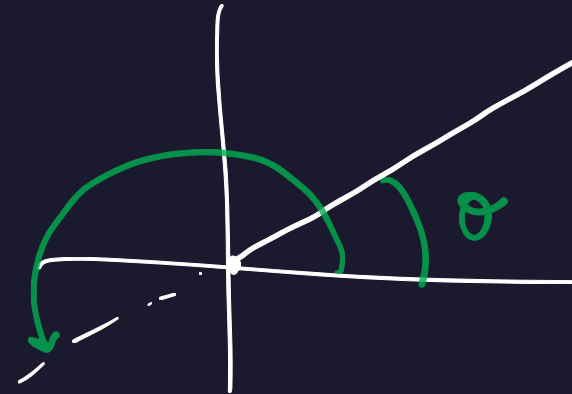
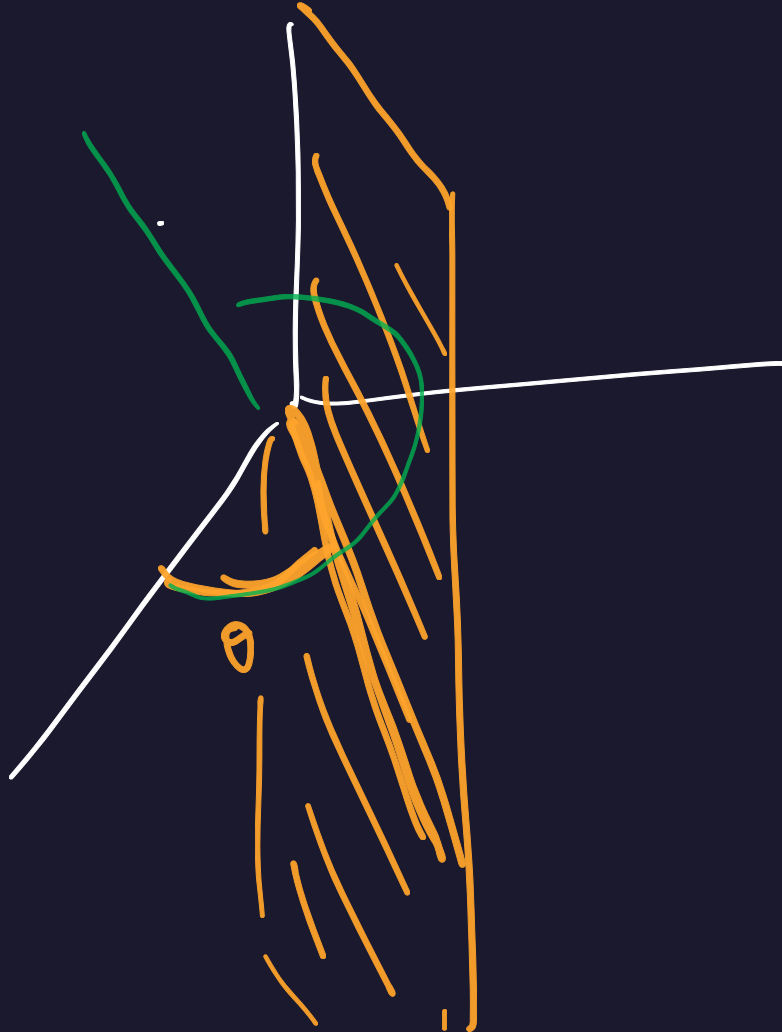
$$\rho^2 = x^2 + y^2 + z^2$$

Example: Sketch the surface given by $\rho = k$. ($k \geq 0$)



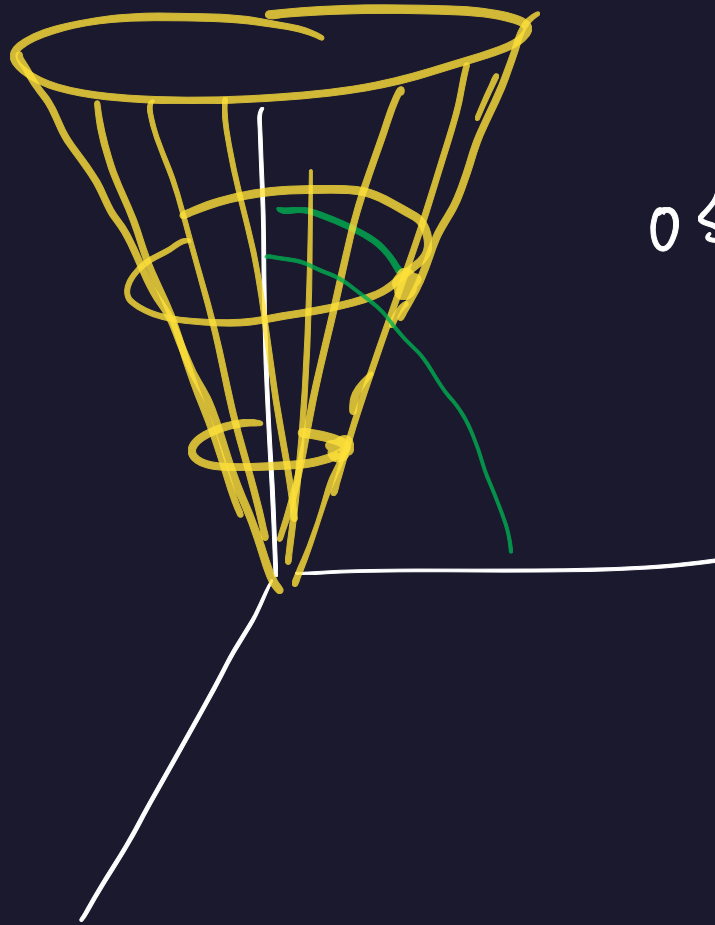
Example: Sketch the surface given by $\theta = k$.

$$\begin{pmatrix} 0 \leq \theta \leq 2\pi \\ 0 \leq k \leq 2\pi \end{pmatrix}$$

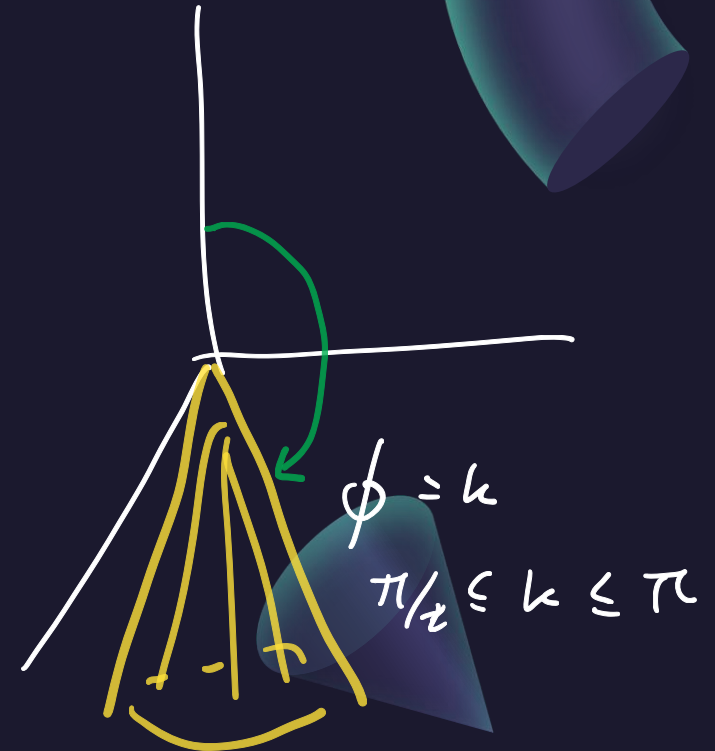


Example: Sketch the surface given by $\phi = k$.

$$(0 \leq \phi \leq \pi)$$



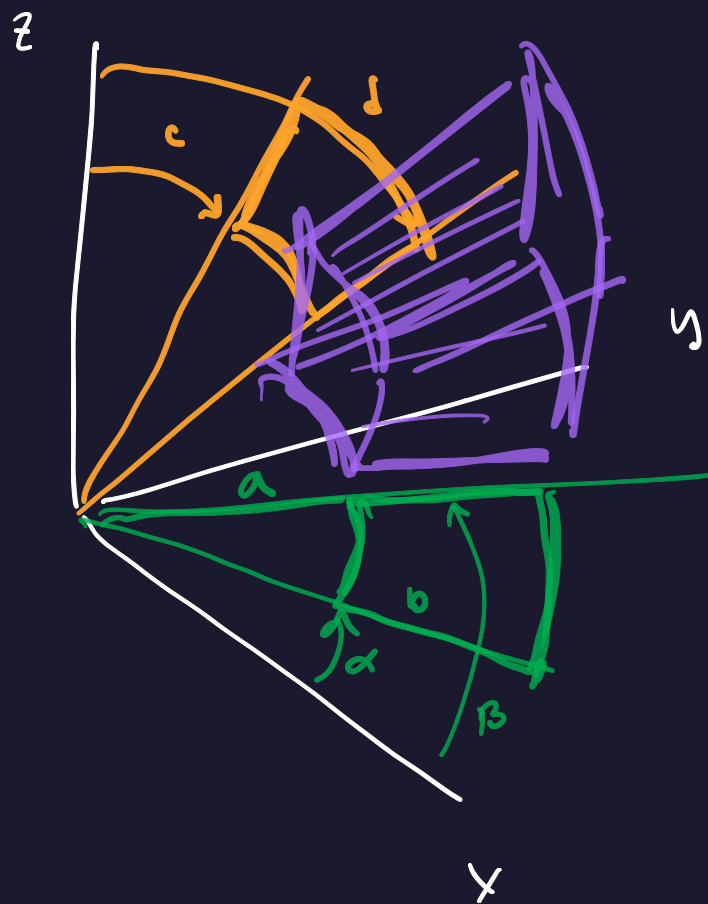
$$0 \leq \phi \leq \pi/2$$



$$\phi = k$$

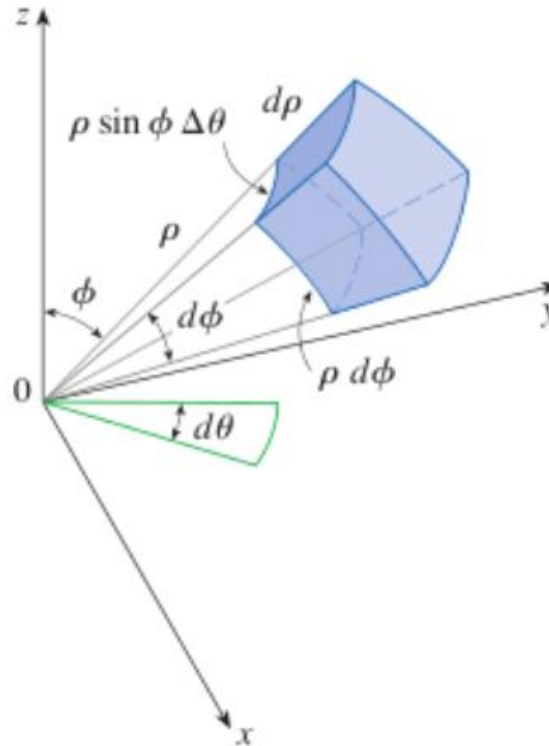
$$\pi/2 \leq k \leq \pi$$

Example: Sketch the “spherical wedge” given by
 $a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d$.



Triple Integrals in Spherical Coordinates

Volume element in spherical coordinates: $dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$



Triple Integrals in Spherical Coordinates

$$\iiint_E f(x, y, z) \, dV$$

$$= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

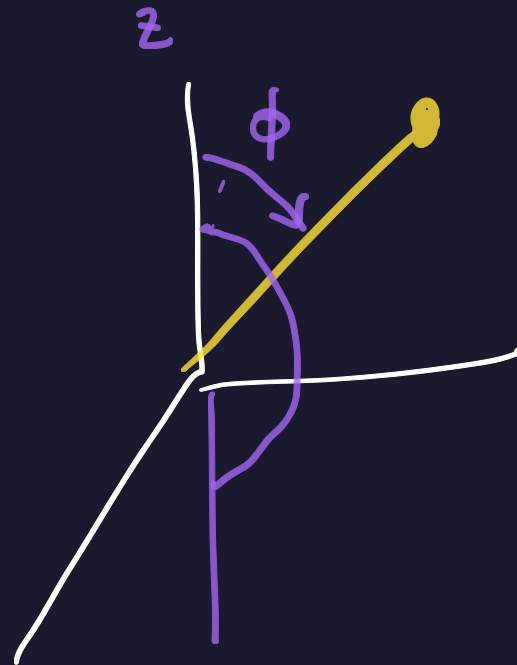
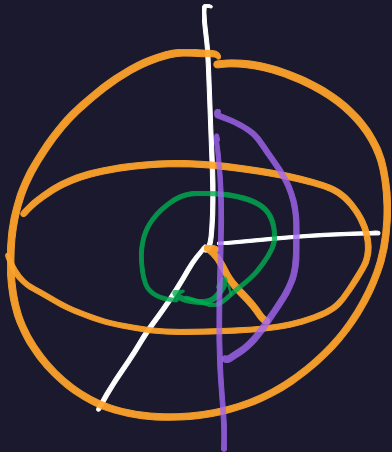
conv. factor!

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

Example: Use triple integrals to find the volume of the sphere of radius R .

$$S = \{(\rho, \theta, \phi) : 0 \leq \rho \leq R, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi\}$$



$$\begin{aligned} & \iiint_S 1 \, dV \\ &= \int_0^\pi \int_0^{2\pi} \int_0^R \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \end{aligned}$$



Example: Use triple integrals to find the volume of the sphere of radius R .

$$\int_0^\pi \int_0^{2\pi} \int_0^R \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi = \int_0^\pi \int_0^{2\pi} \left. \frac{\rho^3}{3} \sin \phi \right|_0^R d\theta \, d\phi$$

$$= \int_0^\pi \int_0^{2\pi} \frac{R^3}{3} \sin \phi \, d\theta \, d\phi = \int_0^\pi \frac{2\pi R^3}{3} \sin \phi \, d\phi$$

$$= -\frac{2\pi R^3}{3} \cos \phi \Big|_0^\pi = \frac{2\pi R^3}{3} + \frac{2\pi R^3}{3} = \frac{4\pi R^3}{3}$$

Example: Evaluate

$$\iiint_E e^{\overbrace{(x^2+y^2+z^2)}^{(\rho^2)^{3/2}} = \rho^3} dV$$

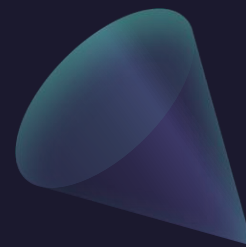
where E is the unit sphere/ball.

$$E = \{ 0 \leq \rho \leq 1, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi \}$$

$$\int_0^\pi \int_0^{2\pi} \int_0^1 e^{\rho^3} \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \quad \downarrow \text{u-sub}$$

$$= \int_0^\pi \int_0^{2\pi} \sin \phi \left. \frac{e^{\rho^3}}{3} \right|_0^1 d\theta \, d\phi = \int_0^\pi \int_0^{2\pi} \sin \phi \left(\frac{e}{3} - \frac{1}{3} \right) d\theta \, d\phi$$

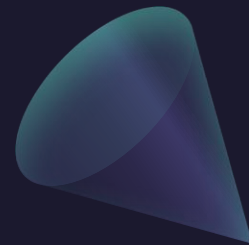
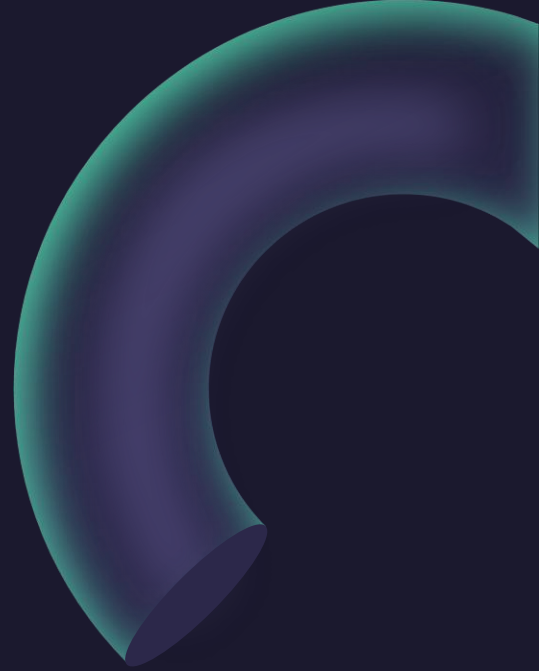
$$= \dots = \boxed{\frac{4}{3}\pi(e-1)}$$



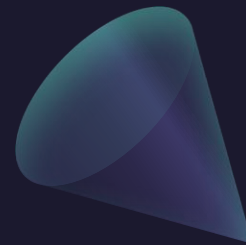
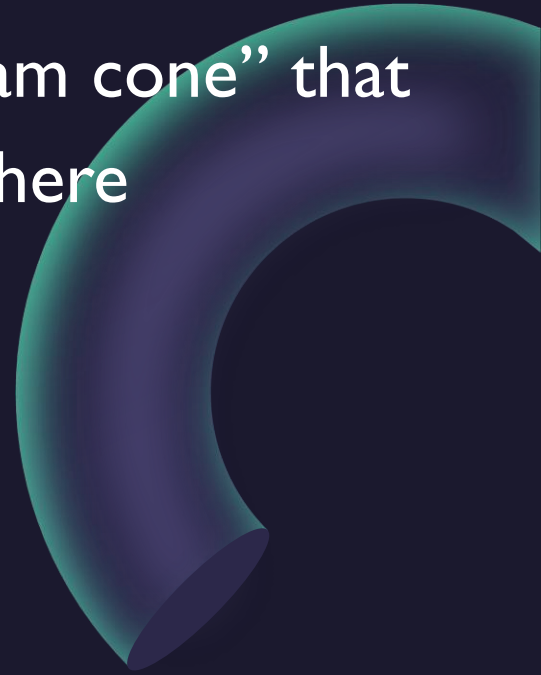
Example: Evaluate

$$\iiint_E e^{(x^2+y^2+z^2)^{3/2}} dV$$

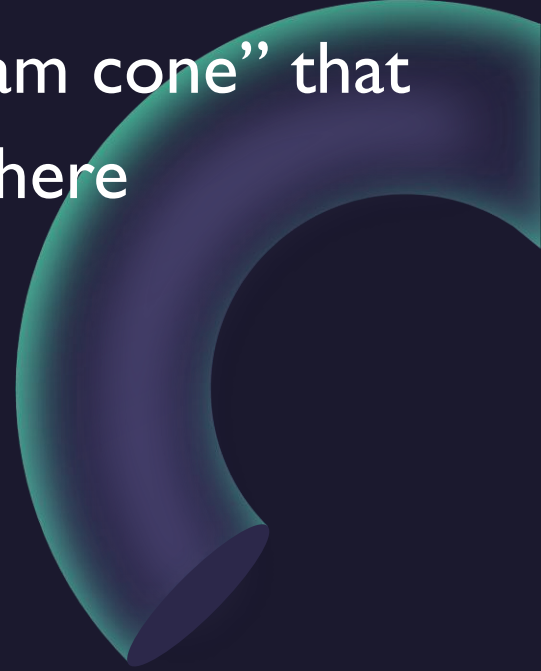
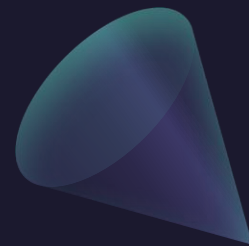
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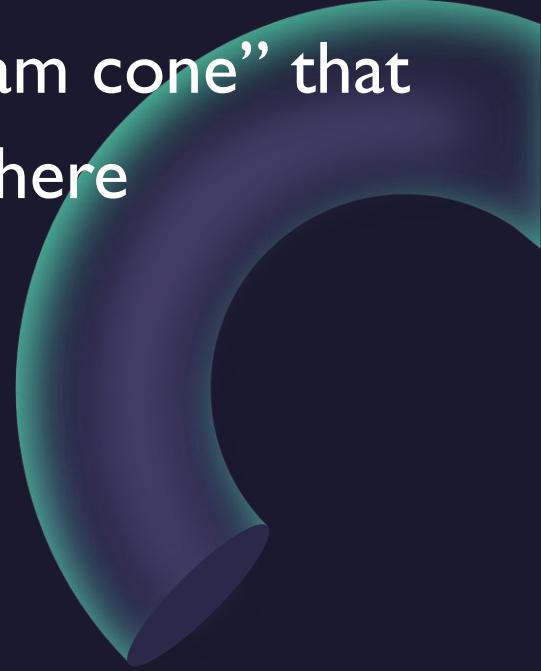
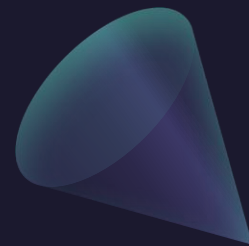
Example: Use triple integrals to find the volume of the “ice-cream cone” that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.



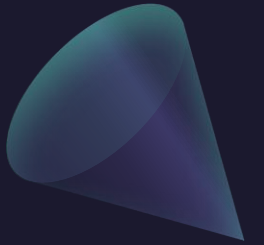
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Questions?



Thank you

Until next time.

